

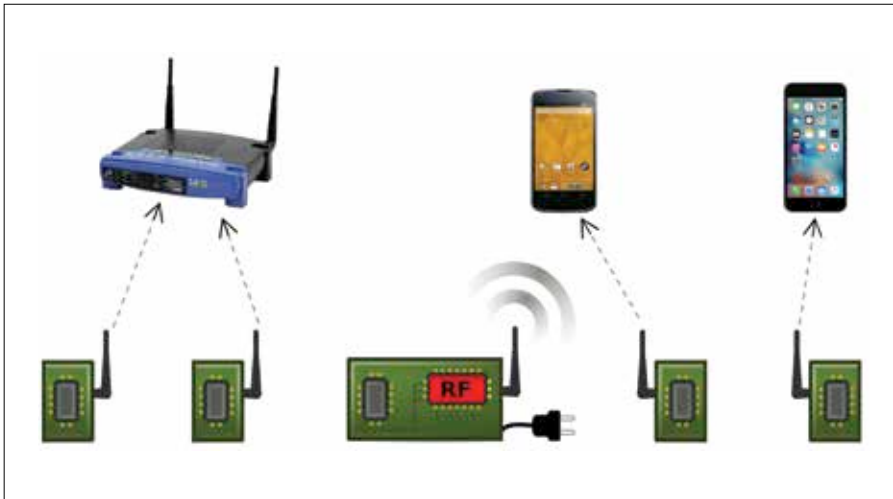
Product update

Apple indeed launched a new phone and tablet but some others are missing the love like the Macbook Pro, one of Apple's core business entities. <http://dgit.in/UpdateReq>



iMessage security

While Apple is on the battlegrounds with the FBI regarding its security concern, researchers found a flaw in its iMessaging service. <http://dgit.in/FlawMsg>



How Passive Wi-Fi will work

due to the starkly different power consumption levels of the digital baseband in comparison to the analog RF – where the digital baseband consumes 10 micro-watts (μ W), the analog RF can consume over 100+ milli-watts (mW) of power. And since the analog RF component in Wi-Fi chipsets is integral for transmitting and receiving data packets, you have to stick with it. Without it, your phone, router, smartwatch, laptop, etc., simply cannot communicate with each other.

In short, analog RF seems to be indispensable for wireless communication to take place. But that's the point – it's not indispensable. That's the whole premise on which Passive Wi-Fi is based on, as it tries to re-imagine wireless radio communications without a critical analog component in it. That's the gamechanger, right there!

Passive Wi-Fi: How it works

What Passive Wi-Fi proposes to do is to have the analog RF component present along with the digital baseband in just a single plugged-in device – like a wireless router or laptop – in any wireless network. Power draw here isn't an issue because the device is plugged into a power source, obviously. For all other non-plugged in, battery operated wireless clients connecting to the wireless network – things smartphones, smartwatches, iPod, etc. – you can have Passive Wi-Fi chips that don't have the analog RF component in them.

How do these Passive Wi-Fi clients communicate back and forth? Since most popular Wi-Fi networks operate at 2.4GHz or 5GHz frequency, the router produces the single radio frequency required for establishing communication between itself and the Passive Wi-Fi devices. That frequency is transmitted to your (Passive) Wi-Fi-enabled device

“Our first offering will be a development kit which allows product developers and engineers to try out Jeeva's exceptionally-low power WiFi modules in your prototypes. These modules allow your device to communicate with a WiFi network for a fraction of the power cost of the best available conventional WiFi transceivers. We tentatively expect to be selling development kits by the second half of 2016.”

AARON PARKS

CO-FOUNDER, JEEVA WIRELESS

through passive sensors that have only the baseband chip. Those Passive Wi-Fi devices not only pick up the radio signal but also mirror it around them, sending forward Wi-Fi packets to any device that has a Wi-Fi chip in it. If this sounds confusing, just watch this video: <http://dgit.in/PassiveWiFiVid>

In a comment published in Wired, Vamsi Talla, who's working on the Passive Wi-Fi project, explained: “The low power passive device isn't transmitting anything at all. It's creating Wi-Fi packets just by reflection. It's a transmission technique that's ultra low-powered, as opposed to a network device.”

Experimental evaluation of Passive Wi-Fi so far claim that “transmissions can be decoded on off-the-shelf smartphones and Wi-Fi chipsets over distances of 30-100 feet in various line-of-sight and through-the-wall scenarios,” according to the University of Washington's website. The researchers also designed a Passive Wi-Fi IC to evaluate power draw of these wireless data signals. They found that roughly 1 and 11 Mbps transmissions consumed 14.48 and 49.28 μ W of power, respectively.

This is big news, because transmitting radio waves at such ridiculously low power consumption levels is tough. Technically, the principle that Passive Wi-Fi uses to create a Wi-Fi network across several devices at such low power consumption levels is called backscattering, which is different from a more traditional mesh network (deployed in ZigBee, Bluetooth LE, etc). In backscattering, nodes blindly transmit data whenever they sense it – Passive Wi-Fi just implements backscattering at never-before-seen levels of power efficiency.

How is Passive Wi-Fi going to change the world?

While Passive Wi-Fi is exciting and promises to revolutionize wireless communications technology, we mustn't get too ahead of ourselves just yet. Remember, for most parts, whatever we've seen of Passive Wi-Fi still remains a tightly controlled research experiment.

There are several hurdles to cross, both with respect to implementation